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Research Article

Data-driven multivariate regression-based anomaly detection and recovery of unmanned aerial vehicle flight data

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Abstract

Flight data anomaly detection is crucial for ensuring the safe operation of unmanned aerial vehicles (UAVs) and has been extensively studied. However, the accurate modeling and analysis of flight data is challenging due to the influence of random noise. Meanwhile, existing methods are often inadequate in parameter selection and feature extraction when dealing with large-scale and high-dimensional flight data. This paper proposes a data-driven multivariate regression-based framework considering spatio-temporal correlation for UAV flight data anomaly detection and recovery, which integrates the techniques of correlation analysis (CA), one-dimensional convolutional neural network and long short-term memory (1D CNN-LSTM), and error filtering (EF), named CA-1DCL-EF. Specifically, CA is first performed on original UAV flight data to select parameters with correlation to reduce the model input and avoid the negative impact of irrelevant parameters on the model. Next, a regression model based on 1D CNN-LSTM is designed to fully extract the spatio-temporal features of UAV flight data and realize parameter mapping. Then, to overcome the effect of random noise, a filtering technique is introduced to smooth the errors to improve the anomaly detection performance. Finally, two common anomaly types are injected into real UAV flight datasets to verify the effectiveness of the proposed method.

Keywords: unmanned aerial vehicle, anomaly detection, data recovery, multivariate regression, error filtering, correlation analysis

1. Introduction

Unmanned aerial vehicles (UAVs) have expanded from military to civilian applications (Liang *et al.*, 2022), such as traffic (Outay *et al.*, 2020), agriculture (Liu *et al.*, 2021; Maes *et al.*, 2019), building inspection and surveillance (Jing & Shimada, 2017), and disaster rescue (Hildmann & Kovacs, 2019; Wang *et al.*, 2022). With the increasing application areas, the global UAV market is projected to reach USD 45.8 billion by 2025 and USD 56.18 billion by 2027 (Ahmad *et al.*, 2022). Despite their promising potential, UAVs face inherent challenges contributing to a higher accident rate than manned aircraft (K. He *et al.*, 2022; Wang *et al.*, 2020; Yang *et al.*, 2023; Zhai & Ye, 2020). Issues like limited real-time control, size constraints, and development costs may lead to severe damage or loss. The safety and reliability of UAVs have been the subject of extensive research and remain a paramount concern (Chen *et al.*, 2022). Flight data play a crucial role in understanding the status and performance of UAVs. Thus, flight data anomaly detection is crucial for enhancing the safety and reliability of UAVs and fostering the advancement and widespread adoption of UAV technology.

UAV flight data anomaly detection approaches mainly include knowledge-based, model-based, and data-driven methods (Yang *et al.*, 2023; Zhong *et al.*, 2022). Knowledge-based methods utilize predefined characteristics and rules of normal UAV flight data to identify anomalies in new flight data, including hidden anomalies that may be challenging to detect using other methods (Theissler, 2017). However, the construction of models or rule bases for normal data in knowledge-based methods relies heavily on expert knowledge, which is limited by the expertise and knowledge level of the experts. Hence, these methods have been primarily utilized in early research (Bu *et al.*, 2017; Qi *et al.*, 2007). Model-based approaches utilize mathematical models to describe the differences between normal and abnormal behaviors of UAVs and employ machine learning algorithms to learn and identify these differences (Freeman *et al.*, 2013). These methods involve training the model with historical flight data to learn the features of normal behavior. Subsequently, new flight data are inputted into the model for prediction, and anomalies are determined based on the prediction errors (Abbaspour *et al.*, 2017; López-Estrada *et al.*, 2016). Despite the advantages of model-based approaches in interpretability, computational cost, and reliability, they encounter challenges in accurately constructing mathematical models due to the complexity of the UAV system. These challenges may lead to model mismatches, resulting in high error rates and limited generalization ability (Liang *et al.*, 2022; Wang *et al.*, 2019).

Unlike knowledge- and model-based approaches, data-driven approaches utilize massive UAV flight data to effectively learn and detect anomalies by employing deep learning or machine learning techniques without relying on expert knowledge or establishing complex physical models. Commonly used deep learning-based

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anomaly detection methods mainly include long short-term memory (LSTM) network (Gu *et al.*, 2022; Jia *et al.*, 2023) or its combination with other models, such as autoencoder (AE, Bae & Joe, 2020; Yang *et al.*, 2023) and principle component analysis (PCA, Wang *et al.*, 2020) to improve the accurate identification and modeling capability of anomalous behavior of UAV flight data. Yang *et al.* (2023) proposed an unsupervised LSTM-AE data-driven method for accurate anomaly detection of a subset of UAV height classes. Bae and Joe (2020) utilized LSTM-AE and AE models with good anomaly detection performance in UAV environment. Zhong et al. (2022) proposed a spatio-temporal correlation-based LSTM (STC-LSTM) data-driven method to achieve anomaly detection of UAV flight data with multiple anomaly types. Wang *et al.* (2019) applied the LSTM with residual filtering method to the anomaly detection of UAV flight data and verified the effectiveness of their method using roll rate as the detected parameter. Wang *et al.* (2020) proposed a PCA-LSTM-based model to realize real-time fault detection of UAVs under on-board constraints. In addition, there are some machine learning-based anomaly detection methods, such as kernel principle component analysis (KPCA, Duan *et al.*, 2017), PCA (Alos *et al.*, 2020), relevance vector machine (RVM, Wang *et al.*, 2019), *k*-nearest neighbor (KNN, Alos & Dahrouj, 2020; Liu & Ding, 2015), and support vector machine (SVM, Bronz *et al.*, 2020; Pan *et al.*, 2020). Duan *et al.* (2017) proposed a KPCA-based data-driven method and obtained satisfactory performance on simulated UAV sensor data. Alos *et al.* (2020) used PCA to detect anomalies in anomalous flights without false alarms. Alos and Dahrouj (2020) proposed a data-driven approach for detecting contextual faults, which was centered on the use of dynamic linear regression to estimate the value of the focus attribute and classifying anomalous data based on KNN. Liu and Ding (2015) first indexed the historical flight data through *k*-dimensional (KD) tree, and then utilized KNN to identify anomalous data and provide reasonable prediction values. Bronz *et al.* (2020) used SVM to construct optimal hyperplanes in high-dimensional spaces to classify abnormal and normal data. Pan *et al.* (2020) proposed an enhanced semi-supervised SVM method integrating active learning by comparing predicted values and uncertainty intervals for anomaly detection in UAV flight data.

The aforementioned publications highlight that researchers have made significant progress in utilizing data-driven methods for anomaly detection in UAV flight data. However, there are still several shortcomings and challenges:

(i) Most existing deep and machine learning-based UAV flight data anomaly detection methods often lack effective parameter selection, which may limit their practical applicability (Bae *et al.*, 2020; Jia *et al.*, 2023). The reason for this limitation is that some unrelated flight parameters may not positively affect the model. While PCC-based parameter selection method may lose the beneficial effects of key nonlinear parameters on the model (Alos *et al.*, 2020; Wang *et al.*, 2019).
(ii) Current methods are mainly based on LSTM for modeling flight data. However, for complex data with spatio-temporal characteristics and multiparameter interactions like UAV flight data, relying on LSTM alone to extract features may have some limitations (Gu *et al.*, 2022; Jia *et al.*, 2023; Wang *et al.*, 2019; Zhong *et al.*, 2022).
(iii) Sensor errors, external environment, and other factors cause random noise in the real UAV flight data. Methods like STC-LSTM (Zhong *et al.*, 2022) and RVM (Wang *et al.*, 2019) may lead to higher false detection rates, as they do not account for random noise.
(iv) Methods like SVM (Bronz *et al.*, 2020), KPCA (Duan *et al.*, 2017), and PCA (Alos *et al.*, 2020) demonstrate relatively good anomaly detection performance, they often overlook the crucial aspect of data recovery.

To address the aforementioned issues and challenges, this paper proposes a data-driven multivariate regression-based framework CA-1DCL-EF for UAV flight data anomaly detection and recovery. First, a correlation analysis (CA) method is introduced to select flight parameters with correlation as model input, mitigating the adverse effects of irrelevant parameters on model performance. Second, by fully utilizing the advantages of one-dimensional convolutional neural network (1D CNN) and LSTM in local feature extraction and capturing time-dependence, a regression model based on 1D CNN-LSTM is designed to capture the spatio-temporal features of the data comprehensively. Finally, to reduce the impact of random noise, an error filtering (EF) technique is introduced to smooth the errors and improve the model's anomaly detection performance. Specifically, compared with previous studies, this paper makes new contributions in the following aspects:

(i) As an alternative to existing methods, a multivariate regression model based on 1D CNN-LSTM is designed to fully extract the spatio-temporal features of UAV flight data by combining the advantages of 1D CNN in local feature extraction and LSTM in temporal feature extraction.
(ii) A novel CA-1DCL-EF framework for UAV flight data anomaly detection and recovery is proposed, which includes CA, EF, anomaly detection, and data recovery, and provides a more comprehensive solution to the practical problems faced in UAV anomaly detection research.
(iii) Experiments on real high-dimensional UAV flight datasets and detailed evaluations and comparisons with baseline methods are conducted; thus, the effectiveness of CA-1DCL-EF is verified.

The rest of this paper is structured as follows. Section 2 provides an overview of the fundamental theory. Section 3 details the proposed framework, outlining its key components and methodology. Section 4 presents the experimental results and analysis obtained by applying the framework. Section 5 summarizes this work and discusses possible future research directions.

2. Preliminaries

2.1. Maximal information coefficient

The maximal information coefficient (MIC) is a widely used nonparametric method for quantifying the correlation between two variables (Reshef *et al.*, 2011). It measures the correlation between variables without making assumptions about specific functional forms or linear relationships. MIC values range from 0 to 1 and demonstrate symmetry. Larger MIC values indicate stronger correlations between the variables. When the MIC value is equal to 0, it indicates that the variables are independent of each other. Given a grid G of size $x \times y$ and a finite set of ordered pairs $(A, B) = (\{a_i, b_i\}, i = 1, 2, \ldots, n)$, n is the variable length. When calculating MIC, mutual information (MI) needs to be calculated first, which is defined as

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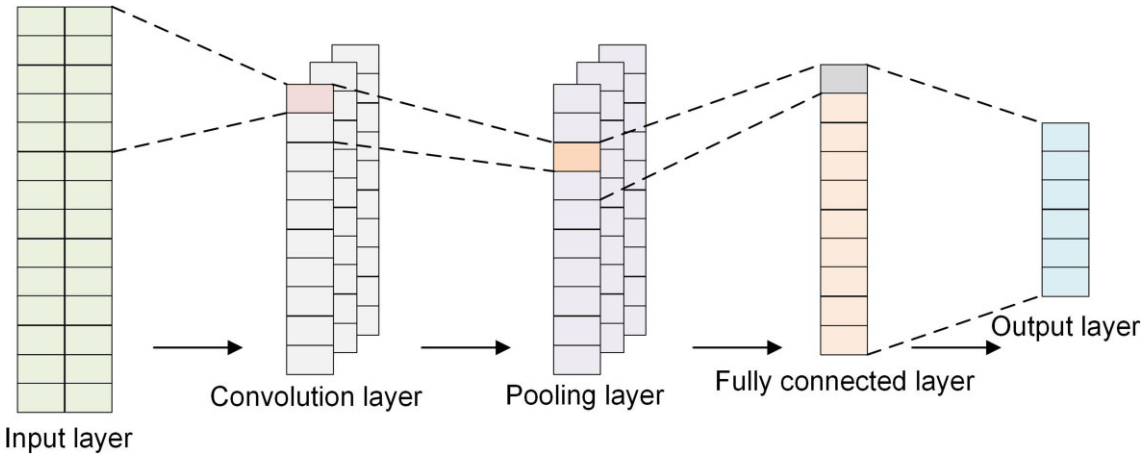

Figure 1: 1D CNN structure diagram.

follows:

$$MI_{(A,B)} = \sum_{a \in A} \sum_{b \in B} p(a,b) \lg \frac{p(a,b)}{p(a)p(b)} \tag{1}$$

where $MI_{(A,B)}$ is MI of A and B, $p(a, b)$ signifies the joint probability density of A and B, and $p(a)$ and $p(b)$ refer to the marginal probability densities of A and B, respectively. The core principle of MIC entails measuring the correlation between two variables by maximizing MI, which is defined as follows:

$$MI^*(D, x, y) = \max MI(D|G) \tag{2}$$

where $MI^*(D, x, y)$ denotes the maximum MI of D on the grid G. $D|G$ indicates that D is being divided using the grid G. D is the data, here it denotes (A, B). In this process, there is involvement with the feature matrix concerning D, enabling MIC to assess MI between variables in different grid partitions to determine the maximum MI value. The feature matrix is defined as follows:

$$MI(D)_{x,y} = \frac{MI^*(D, x, y)}{\log \min\{x, y\}} \tag{3}$$

where $\log \min\{x, y\}$ is the element of the normalized feature matrix. Therefore, based on equation (3), the MIC can be defined as follows:

$$MIC(D)_{x,y} = \max_{xy<C(n)} \{M(D)_{x,y}\} \tag{4}$$

where $C(n)$ is the maximum value of the grid G and $w(1) \leq C(n) \leq O(n^{1-\varepsilon})$, $0<\varepsilon<1$. ε is a positive number less than 1, used to adjust the upper limit of $C(n)$. $w(1)$ is a parameter related to the sample size and is typically a small positive number.

2.2. One-dimensional convolutional neural network

The 1D CNN (Kiranyaz *et al.*, 2021; Li *et al.*, 2022) is utilized to extract features from different positions of the input signal using sliding windows. It typically consists of convolutional, pooling, and fully connected layers, as shown in Fig. 1.

The convolution layer is a fundamental component of the 1D CNN architecture, extracting local features from the input data. Its calculation is shown as follows:

$$x_j^l = g\left(\sum_{i \in Fj} x_i^{l-1} \cdot k_{ij}^l + b_j^l\right) \tag{5}$$

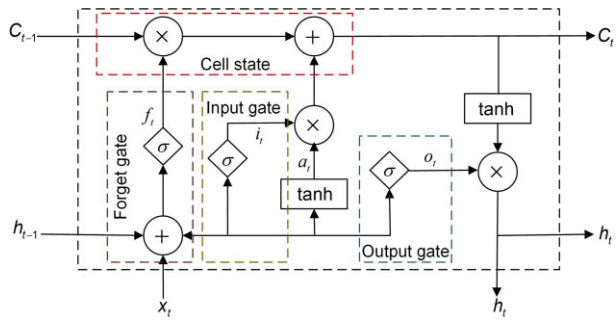

Figure 2: LSTM structure diagram.

where x_j^{l+1} is the jth element of the in the lth layer of the input feature F_j. k, b, and $g(\cdot)$ are the convolution kernel, bias, and the nonlinear activation function of the convolution layer, respectively. The pooling layer plays a crucial role in reducing the number of parameters in the model while preserving essential features and enhancing its generalization capability, which is defined as follows:

$$x_j^{l+1} = Pooling(x_j^l) \tag{6}$$

where x_j^{l+1} denotes the jth element of the $l + 1$th layer and $Pooling(\cdot)$ is the pooling operation. The fully connected layer is computed as follows:

$$x^l = f(w^l x^{l-1} + b^l) \tag{7}$$

where ω^l, b^l, and $f(\cdot)$ are the weights, bias, and nonlinear activation functions of the fully connected layer, respectively.

2.3. Long short-term memory

LSTM is a variant of recurrent neural networks that excels at capturing long-term dependencies (Graves, 2012). There are forget gate f_t, input gate i_t, output gate o_t, and memory cell of LSTM, as shown in Fig. 2. f_t is the initial recipient of information, determining which piece of historical information to retain or forget in the current state. Suppose the input data are x_t at time t, f_t is defined as follows:

$$f_t = \sigma(w_{hf} h_{t-1} + w_{xf} x_t + b_f) \tag{8}$$

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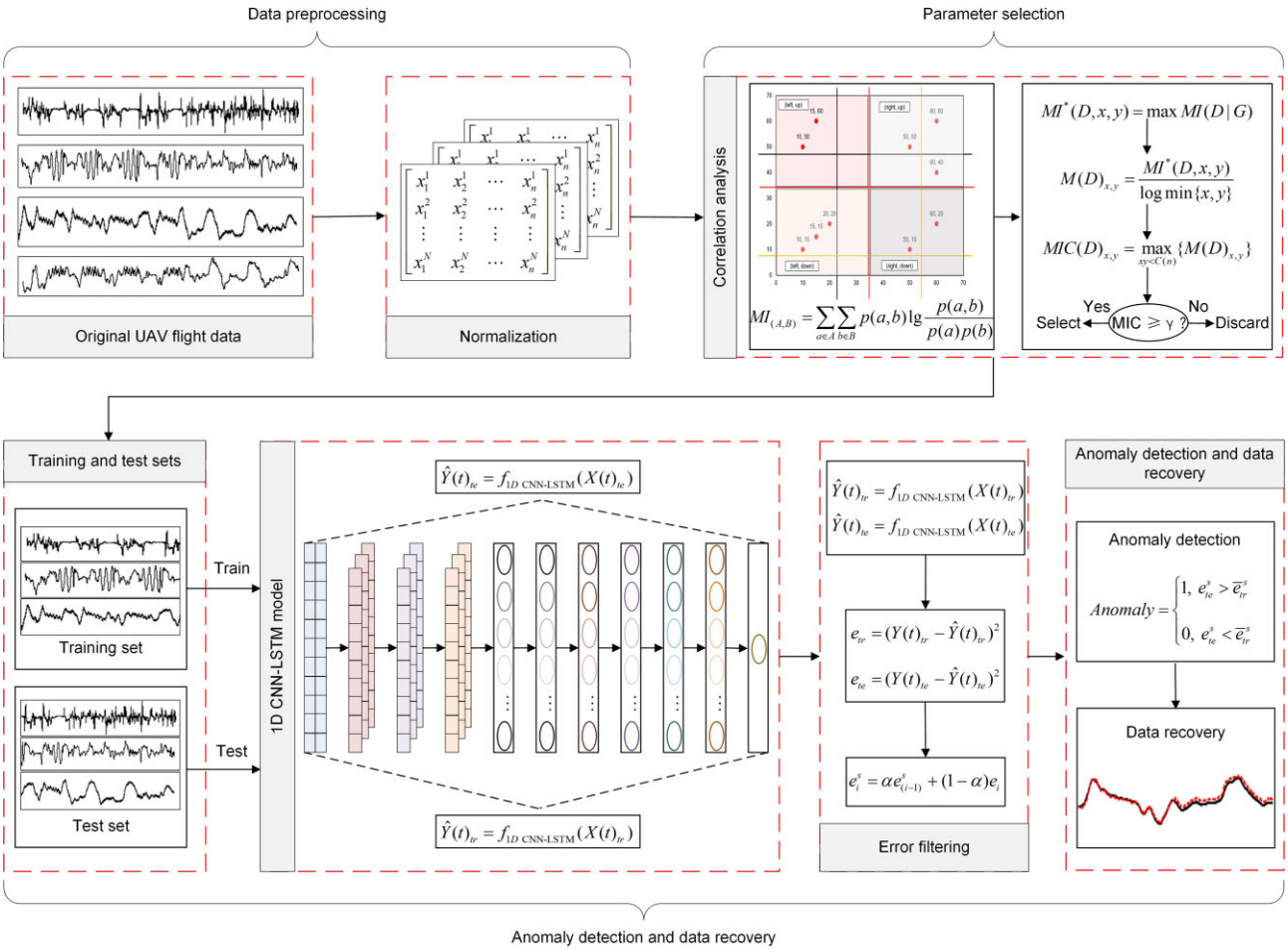

Figure 3: The proposed CA-1DCL-EF framework.

where σ is the activation function, w_{hf} and w_{xf} are the weights of f_t. b_f is the bias of f_t. h_{t-1} denote the cell state and output vector at time $t-1$. i_t is used to control which information from the current input to the network flows into the memory cell, which is defined as follows:

$$i_t = \sigma(w_{hi}h_{t-1} + w_{xi}x_t + b_i) \tag{9}$$

$$\tilde{c}_t = \tanh(w_{xc}x_t + w_{hc}h_{t-1} + b_c) \tag{10}$$

$$c_t = f_t \otimes c_{t-1} + i_t \otimes \tilde{c}_t \tag{11}$$

where w_{hi} and w_{xi} are the weights of i_t. b_i is the bias of i_t. $\tilde{c}_t$ is the intermediate state parameter used for updating at time t. w_{hc} and w_{xc} are the weights of $\tilde{c}_t$. b_c is the bias of $\tilde{c}_t$. tanh is the activation function. c_{t-1} denotes the output vector at time $t-1$. $\otimes$ denotes the dot product. o_t controls which part of the memory cell will be output at time t. It is defined as follows:

$$o_t = w_{ho}h_{t-1} + w_{xo}x_t + b_o \tag{12}$$

$$h_t = o_t \otimes \tanh(c_t) \tag{13}$$

where w_{ho} and w_{xo} are the weights of o_t. b_o is the bias of o_t.

3. Methodology Framework

This paper proposes a data-driven multivariate regression-based framework CA-1DCL-EF considering spatio-temporal correlation, as shown in Fig. 3. It innovatively integrates CA, 1D CNN-LSTM, and EF. First, the raw UAV flight data are normalized to eliminate the effect of different magnitudes. Second, MIC-based CA is performed to select parameters with correlation, which are then used as inputs to the model. Then, 1D CNN-LSTM model is trained and evaluated using training and test sets. Finally, the training errors are smoothed to obtain the anomaly determination threshold. Anomaly detection is achieved by comparing the smoothed test errors with the threshold, and data recovery is achieved by replacing the anomalous data with the predicted data. Specifically, the proposed framework CA-1DCL-EF mainly includes three parts: data preprocessing, parameter selection, and anomaly detection and recovery, which are described in detail in this section.

3.1. Data preprocessing

The original UAV flight data are normalized using the maximum–minimum normalization method (Jahan & Edwards, 2015). It adjusts each feature range to [0,1] to facilitate the model to learn the features better. It is defined as follows:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{14}$$

where x represents the original data and x' the normalized data. $x_{\max}$ and $x_{\min}$ are the maximum and minimum values of x, respectively.

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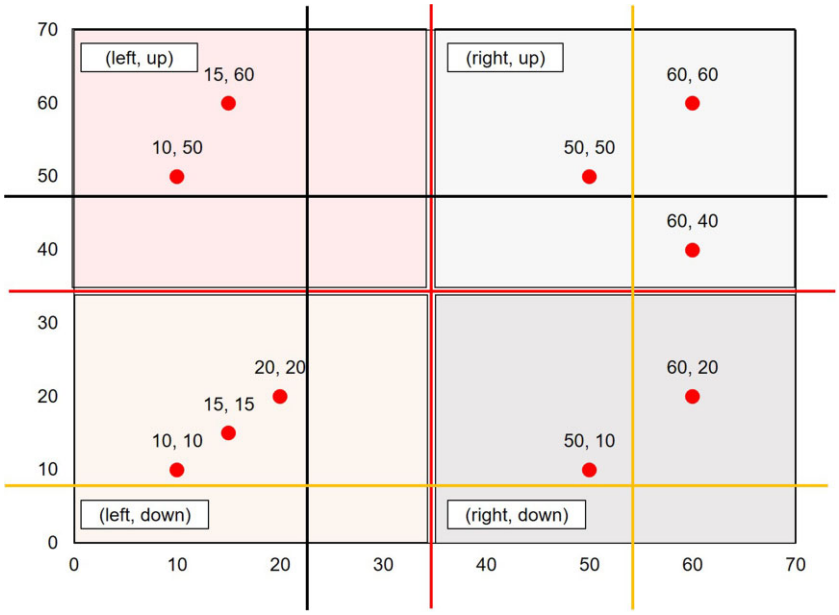

Figure 4: Schematic diagram of different grid division schemes.

3.2. MIC-based parameter selection

Since there are numerous flight parameters, it is necessary to perform parameter selection to reduce the risk of model underfitting. However, linear and nonlinear correlation parameters in the flight data need to be fully considered in the parameter selection process to avoid the loss of critical parameters that may positively affect the model performance. Compared with PCC, MIC can capture both linear and nonlinear correlations between parameters. An illustrative example is presented in Fig. 4, where different-colored lines represent distinct grid division schemes and (A, B) contains 10 ordered pairs. Considering the grid scheme divided by the red lines as an example, the joint and boundary probability densities of variables (A, B) are as follows: $p(\text{left, up}) = 0.2$, $p(\text{right, up}) = 0.3$, $p(\text{left, down}) = 0.3$, $p(\text{right, down}) = 0.2$, $p(\text{left}) = 0.5$, $p(\text{right}) = 0.5$, $p(\text{up}) = 0.5$, and $p(\text{down}) = 0.5$. The MI under this scheme can be calculated according to equation (1), i.e.,

$$MI_{(A,B)} = 0.2 * \left(\log\left(\frac{0.2}{0.5 * 0.5}\right) + \log\left(\frac{0.2}{0.5 * 0.5}\right)\right) + 0.3 * \left(\log\left(\frac{0.3}{0.5 * 0.5}\right) + \log\left(\frac{0.3}{0.5 * 0.5}\right)\right) = 0.0087 \tag{15}$$

The MI values of other grid schemes, such as those divided by orange and black lines, are also computed. The grid division scheme with the highest MI value is chosen as the final scheme based on the calculation described in equation (2). Subsequently, the maximum value is normalized using equation (3), resulting in the maximum MI value, denoted as MIC. A correlation between variables A and B exists when MIC is greater than or equal to γ. The calculation of MIC can be performed using the MINE package (Reshef *et al.*, 2012), and its procedure is outlined in Algorithm 1.

Algorithm 1: MIC calculation process.

Input: Multivariate variables with m dimension

S1: Calculate the maximum mutual information:

for each grid G **do**

Partition scatterplot of the variable pairs in each grid G

Calculate $MI(D, x, y)$ of each grid

Calculate $MI^*(D, x, y)$

end for

S2: Calculate the feature matrix $MI(D)_{x,y}$:

Generate the characteristic matrix

S3: Calculate MIC:

Calculate MIC with limited grid size, $C(n)=n^{1-\varepsilon}$

Output: MIC $\in R^{n \times n}$, $n \leq m$

3.3. CA-1DCL-EF-based anomaly detection and recovery

3.3.1. *Model architecture*

Figure 5 shows the structure diagram of the designed 1D CNN-LSTM regression model. 1D CNN captures local features of the input data through convolutional operations, effectively addressing local correlations in the time series and reducing the model's

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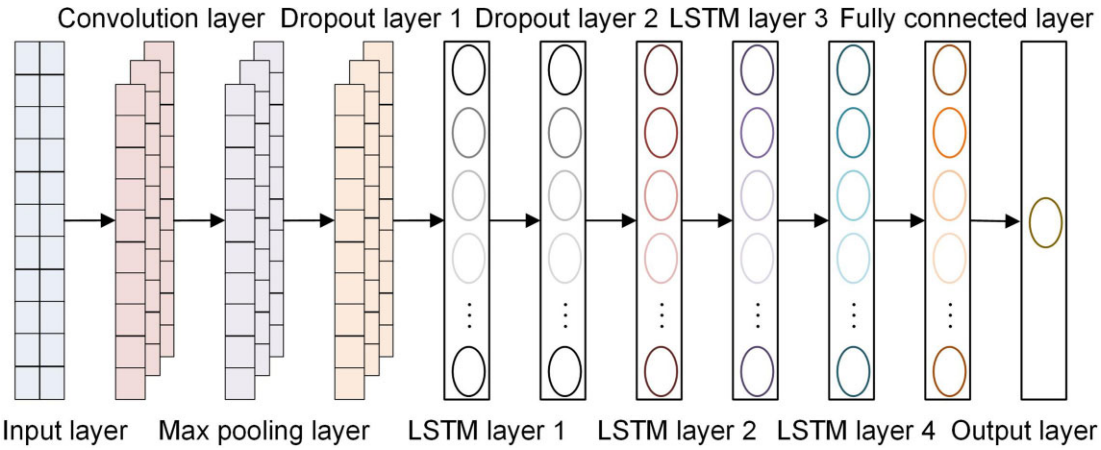

Figure 5: The designed 1D CNN-LSTM structure diagram.

Table 1: Input and output shapes of each layer.

Layer	Input shape	Output shape
Convolution layer	(None, S, *n*)	(None, S, 128)
Max pooling layer	(None, S, 128)	(None, 1, 128)
Dropout layer 1	(None, 1, 128)	(None, 1, 128)
LSTM layer 1	(None, 1, 128)	(None, 1, 64)
Dropout layer 2	(None, 1, 64)	(None, 1, 64)
LSTM layer 2	(None, 1, 64)	(None, 1, 32)
LSTM layer 3	(None, 1, 32)	(None, 1, 16)
LSTM layer 4	(None, 1, 16)	(None, 8)
Fully connected layer	(None, 8)	(None, 1)

parameter count. LSTM utilizes its gate mechanism to regulate the information flow, which can capture long-term dependencies. Combining the strengths of 1D CNN and LSTM, 1D CNN-LSTM can more comprehensively capture spatio-temporal features in flight data and improve the prediction ability of time series data. Specifically, 1D CNN-LSTM contains a convolutional layer, a max pooling layer, two dropout layers, four LSTM layers, and a fully connected layer. The input and output shapes of each layer are shown in Table 1.

3.3.2. Model training

The UAV flight parameters with correlation are first normalized using equation (14) before the model training. Then, the normalized parameters are reconstructed to match the input format of the model, which is defined as follows:

$$X(t) = \begin{bmatrix} x_1^{t-1}, & x_1^{t-2}, \ldots, & x_1^{t-S} \\ x_2^{t-1}, & x_2^{t-2}, \ldots, & x_2^{t-S} \\ \vdots & & \\ x_n^{t-1}, & x_n^{t-2}, \ldots, & x_n^{t-S} \end{bmatrix}, Y(t) = y(t) \tag{16}$$

where $X(t)$ is the model input, S is the sliding window, and n is the number of parameters with correlation, respectively. Figure 6 shows the schematic diagram of the sliding window. Through extensive exploratory experiments, combined with data characterization. In this paper, S was set to 5. $Y(t)$ and $y(t)$ are the target values of the model output and the detected parameter at time t. The predicted value $\hat{Y}(t)$ of the model is obtained after inputting $X(t)$ into the model. The objective of model training is to minimize the error between $\hat{Y}(t)$ and $Y(t)$, which is defined as follows:

$$J = \frac{1}{l}\sum_{i=1}^{l}(\hat{y}_i - y_i)^2 \tag{17}$$

where $\hat{y}_i$ and y_i are the ith elements of $\hat{Y}(t)$ and $Y(t)$, respectively, and l is the sample length.

Two dropout layers are added to mitigate the model overfitting phenomenon. In each training iteration, the dropout layer randomly discards some neurons from the previous layer with a certain probability. This helps to alleviate the complex collaborative relationships between neurons, thus enabling the model to learn more robust features and improving the model's generalization ability (Tan *et al.*, 2018). The training process is shown in Algorithm 2.

Algorithm 2: Model training process.

Input: Data set D with correlation, sliding window S, epochs E, optimizer, batch_size s, learning rate ℓ
S1: Data normalization $D_{nor} \leftarrow x' = (x - x_{min})/(x_{max} - x_{min})$
S2: Train, test=divide (D_{nor}, 0.8)
S3: Training:
while not reach to maximum E **do:**
  for epoch in range (E) **do:**
    for j in range (len(X_{tr})) **do:**
      Input, target=X_{tr}, Y_{tr}
      Mfit=fit(X_{tr}, Y_{tr}, E, s, shuffle=False)
      Compute the loss function $J = 1/l\sum_{i=1}^{l}(\hat{y}_i - y_i)^2$
      Parameter update=Adam optimizer (ℓ)
      Find and save the best model based on J
    end for
  end for
end while
S4: Compute the prediction error $e_i = (\hat{y}_i - y_i)^2$
S5: Smooth error $e_i^s = \alpha e_{i-1}^s + (1\text{-}\alpha)\, e_i$
S6: Compute the anomaly determination threshold $T = 1/l\sum_{i=1}^{l} e_i^s$
Output: Predicted value $\hat{y}_i$, error e_i^s, threshold T

3.3.3. Anomaly detection and data recovery

For the training and test sets $X(t)_{tr}$ and $X(t)_{te}$, the $\hat{Y}(t)_{tr}$ and $\hat{Y}(t)_{te}$ are obtained after the regression of the 1D CNN-LSTM model, which can be defined as follows:

$$\hat{Y}(t)_{tr} = f_{1D\,CNN-LSTM}(X(t)_{tr}) \tag{18}$$

$$\hat{Y}(t)_{te} = f_{1D\,CNN-LSTM}(X(t)_{te}) \tag{19}$$

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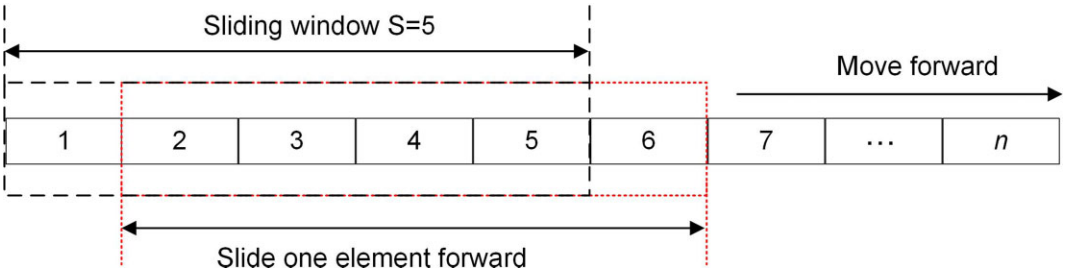

Figure 6: The schematic diagram of the sliding window.

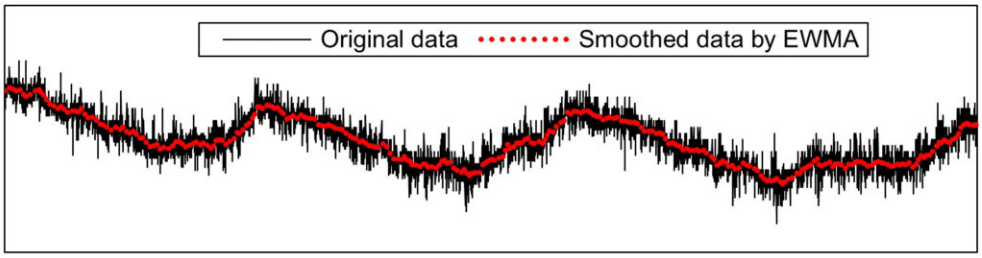

Figure 7: Example of data smoothing using EWMA.

where $\hat{Y}(t)_{tr}$ and $\hat{Y}(t)_{te}$ are the predicted values of the training and test sets, respectively. $f_{1D\ CNN-LSTM}(\cdot)$ is the mapping function. Then, the errors e_{tr} and e_{te} of the training and test sets can be defined as follows:

$$e_{tr} = (Y(t)_{tr} - \hat{Y}(t)_{tr})^2 \tag{20}$$

$$e_{te} = (Y(t)_{te} - \hat{Y}(t)_{te})^2 \tag{21}$$

where $Y(t)_{tr}$ and $Y(t)_{te}$ are the target values of the training and test sets, respectively. To mitigate the impact of random noise in the original UAV flight data, the exponentially weighted moving average (EWMA) method is employed (Lucas & Saccucci, 1990). EWMA applies an exponentially weighted average operation to the original time series data, assigning weights to each data point based on an exponential function. The calculation of EWMA can be defined as follows:

$$e_i^s = \alpha e_{(i-1)}^s + (1-\alpha)e_i \tag{22}$$

where e_i^s is the ith smoothed error value, α is the adjustable weight parameter, and e_i is the ith actual error value. Figure 7 shows an example of data smoothing using EWMA. The points present in the data that change drastically can be filtered out by EWMA and retain the most valuable and critical information.

The e_{tr}^s and e_{te}^s are obtained after smoothing e_{tr} and e_{te}, respectively, using equation (22). Then, anomaly detection can be defined as follows:

$$Anomaly = \begin{cases} 1, & e_{te}^s > \bar{e}_{tr}^s \\ 0, & e_{te}^s < \bar{e}_{tr}^s \end{cases} \tag{23}$$

where $\bar{e}_{tr}^s$ is the average value of e_{tr}^s, 1 indicates abnormal, and 0 indicates normal. For data recovery, the $\hat{Y}(t)_{te}$ shown in equation (19) is the recovery value.

4. Experiments

4.1. Data description

The UAV flight datasets used in this paper were collected by the University of Minnesota (Taylor, 2013, 2014). They collected flight data from various types of UAVs, and data from multiple flights performed by the same UAV type. These datasets have been widely used in research on the anomaly detection of UAV flight data (Y. He *et al.*, 2017; Pan, 2017, 2018; Yang *et al.*, 2023). Specifically, the Thor Flight 98 and Thor Flight 104 datasets used in this study are recorded data from the 98th and 104th flights of the Thor UAV. Each dataset comprises 84 parameters sampled at a frequency of 50 Hz. The length of each parameter in the Thor Flight 98 dataset and Thor Flight 104 dataset is 21 015 and 25 836, respectively. This paper uses the flight data from the Thor UAV during level flight with 16 000 data sampling points.

Since anomaly data are difficult to obtain, this paper uses anomaly injection method to generate anomaly data. Specifically, two common types of anomalies in UAVs are considered: bias and drift anomalies, shown in equations (24) and (25), respectively (Freeman *et al.*, 2013; Qi *et al.*, 2013). The bias anomaly occurs when the UAV fails to maintain the expected trajectory, position, or attitude due to fixed factors, such as sensor errors, environmental disturbances, or power system issues, making the UAV flight data deviate consistently from the anticipated values. The drift anomaly involves a gradual deviation of UAV flight data from the intended trajectory or position, typically caused by dynamic factors, such as control system instability, attitude inaccuracies, or variations in wind speed.

$$y(t)_{bias} = y(t) + \mu \tag{24}$$

where $y(t)$ are the original flight data, μ is a constant, and t is a period of time.

$$y(t)_{drift} = y(t) + \theta(t) \tag{25}$$

where $\theta(t)$ is a constant or a function of a period of time t.

NAV north velocity (NAVVN) is a crucial UAV parameter that can improve the accuracy and reliability of UAV navigation. Therefore, this paper selects NAVVN as the detected parameter to verify the model's performance. Figure 8 illustrates the flight datasets with injected bias and drift anomalies, and the data division. The anomaly injection range is [14 400:16 000]. For the bias anomaly,

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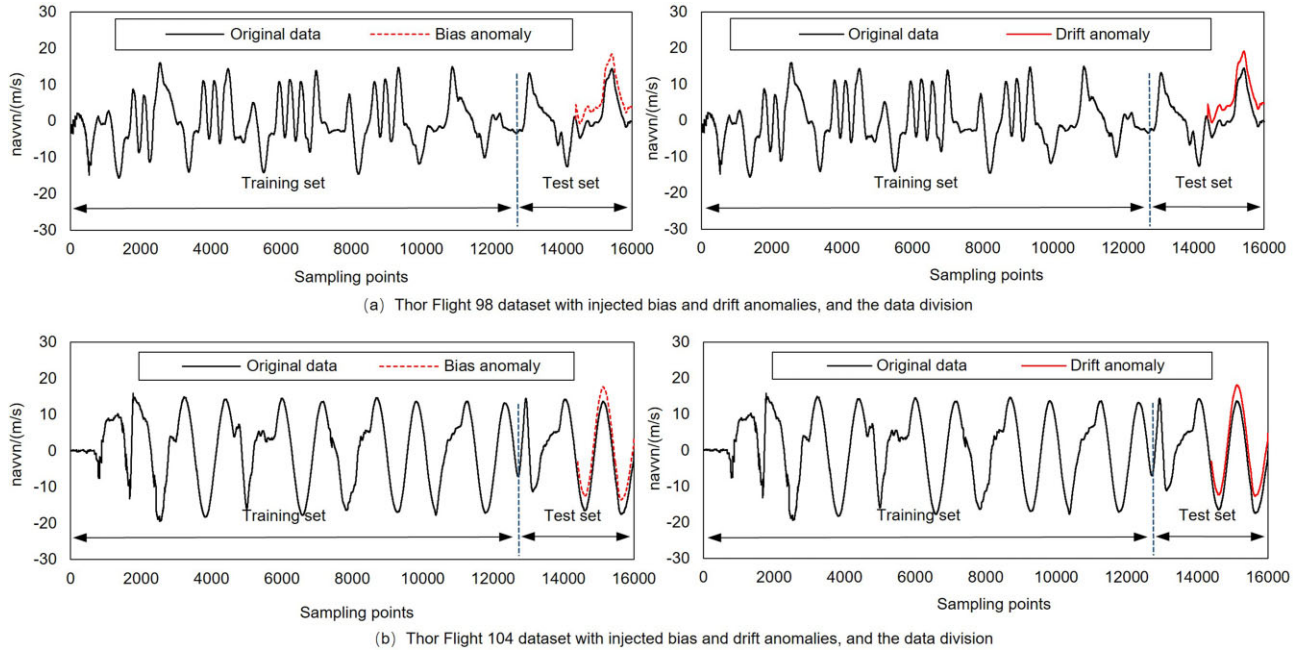

Figure 8: Flight datasets with injected bias and drift anomalies, and the data division.

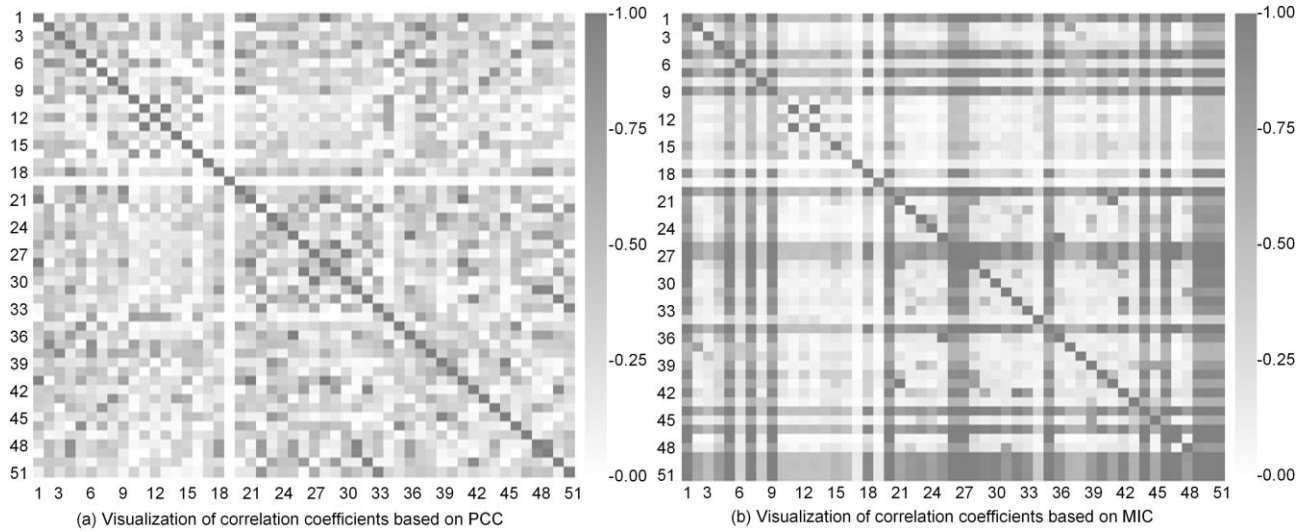

Figure 9: Visualization of correlation coefficient results based on PCC and MIC for Thor Flight 98 dataset.

the value of μ is set to 4. Regarding the drift anomaly, 1600 points are evenly sampled from the range [4,5] using the linespace function and added to the corresponding points within the interval [14 400:16 000]. The training set comprises the range [0:12 800], which accounts for 80% of the dataset, while the test set consists of the range [12 800:16 000], representing 20% of the dataset.

4.2. Experimental set-up

In the experiments, the operating system is Windows 10 (64-bit) and an AMD Ryzen 5 3600 6-Core @3.6 GHz processor is utilized for deep learning tasks. As the deep learning framework, TensorFlow version 2.12.0 is chosen. During model training, the following hyperparameter settings are used: a learning rate of 0.001, a batch size of 128, a training epoch of 100 periods, and a dropout rate of 0.1. For the optimizer, the Adam optimizer is used to tune the model parameters.

4.3. The results of MIC-based parameter selection

After removing redundant parameters, 51 parameters are used for CA. Taking Thor Flight 98 dataset as an example, Fig. 9 shows the visualization of its correlation coefficient results based on MIC and PCC. Darker colors indicate stronger correlations and vice versa, with MIC displaying a greater number of darker grids compared with PCC. Table 2 shows the correlation coefficients of Thor Flight 98 dataset based on MIC and PCC between the detected parameter NAVVN. Although PCC has high correlation coefficients between parameters with strong linear correlation, it is worth not-

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Table 2: Flight parameters with correlation to NAVVN in Thor Flight 98 dataset.

Name	Description	Unit	PCC	MIC
alt	GPS altitude (WGS84)	m	0.02	**0.87**
ax_bias	Accelerometer bias, X-axis	rad/s	0.11	**0.88**
ay_bias	Accelerometer bias, Y-axis	rad/s	0.11	**0.85**
az_bias	Accelerometer bias, Z-axis	rad/s	0.03	**0.86**
GPS_TOW	Time of week from GPS	s	0.10	**0.89**
hx	X-axis magnetic field	Gauss	**0.85**	**0.62**
lat	GPS latitude	deg	0.10	**0.85**
lon	GPS longitude	deg	0.55	**0.88**
psi	Euler roll angle	rad	0.10	**0.80**
p_bias	Gyro bias, X-axis (roll)	rad/s	0.12	**0.88**
q_bias	Gyro bias, Y-axis (pitch)	rad/s	0.10	**0.80**
r_bias	Gyro bias, Z-axis (yaw)	rad/s	0.10	**0.88**
vd	GPS down velocity	m/s	0.36	**0.87**
ve	GPS east velocity	m/s	0.03	**0.88**
vn	GPS north velocity	m/s	**0.95**	**0.89**

ing that more accurately MIC captures the nonlinear correlations between parameters. In this work, the correlation determination threshold γ is set to 0.6, which is derived from previous research and experience (Alos *et al.*, 2020; Yang *et al.*, 2023; Zhong *et al.*, 2022). Based on the MIC results, 15 parameters associated with NAVVN are selected as input parameters for the model. In order to clearly demonstrate the distribution status and probability density of the used UAV flight datasets, the normalized parameter features are visualized using violin plots, as shown in Fig. 10. The original range of values for each parameter feature is shown in the black boxes in Fig. 10.

4.4. The results of CA-1DCL-EF-based anomaly detection and recovery

4.4.1. *Evaluation metrics*

Anomaly detection focuses on the ability of the model to find and identify anomalies. The true positive rate (TPR), false positive rate (FPR), and accuracy (ACC) rate (Miao *et al.*, 2018; Wang *et al.*, 2019) are used to evaluate anomaly detection performance metrics. Data recovery aims to mitigate the effects of anomalies and return to safe flight conditions on time. The mean absolute error (MAE) and mean square error (MSE) are used to evaluate data recovery performance (Hodson *et al.*, 2022; Sun *et al.*, 2019). The larger TPR and ACC values and the smaller FPR value indicate better model anomaly detection performance. The smaller MAE and MSE values indicate better model data recovery performance. Their calculation formulas are shown in equations (26–30):

$$TPR = \frac{TP}{TP + FN} \times 100\% \tag{26}$$

$$FPR = \frac{FP}{TN + FP} \times 100\% \tag{27}$$

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \times 100\% \tag{28}$$

$$MAE = \frac{1}{L}\sum_{i=1}^{L} |y_i - \hat{y}_i| \tag{29}$$

$$MSE = \frac{1}{L}\sum_{i=1}^{L} (y_i - \hat{y}_i)^2 \tag{30}$$

where *TP* represents the number of correctly identified normal samples, *TN* represents the number of correctly identified abnormal samples, *FP* represents the number of normal samples incorrectly identified as abnormal, and *FN* represents the number of abnormal samples incorrectly identified as normal. *L* denotes the length of each parameter, y_i represents the *i*th data point of the original value without anomaly, and $\hat{y}_i$ represents the *i*th data point of the estimated value.

4.4.2. *Baseline methods*

To verify the effectiveness of CA-1DCL-EF, a comparison is conducted with the state-of-the-art regression methods based on LSTM-RF (Wang *et al.*, 2019) and STC-LSTM (Zhong *et al.*, 2022). In addition, for a comprehensive baseline comparison, KNN and support vector regression (SVR) methods are also included. Brief descriptions of LSTM-RF, STC-LSTM, KNN, and SVR are provided as follows:

(i) LSTM-RF and STC-LSTM: STC-LSTM and LSTM-RF are based on LSTM for modeling UAV flight data and achieve anomaly detection by comparing the prediction errors with the anomaly detection threshold. Unlike STC-LSTM, LSTM-RF applies additional smoothing to the prediction errors. The network architecture of STC-LSTM consists of three LSTM layers and two fully connected layers. The structure of LSTM-RF consists of an input layer, an LSTM layer, and an output layer.

(ii) KNN: KNN assesses the degree of anomaly by calculating the distance or density of each data point to its KNNs. Anomalies are usually considered to be points that are farther or sparser relative to their neighbors. The predicted

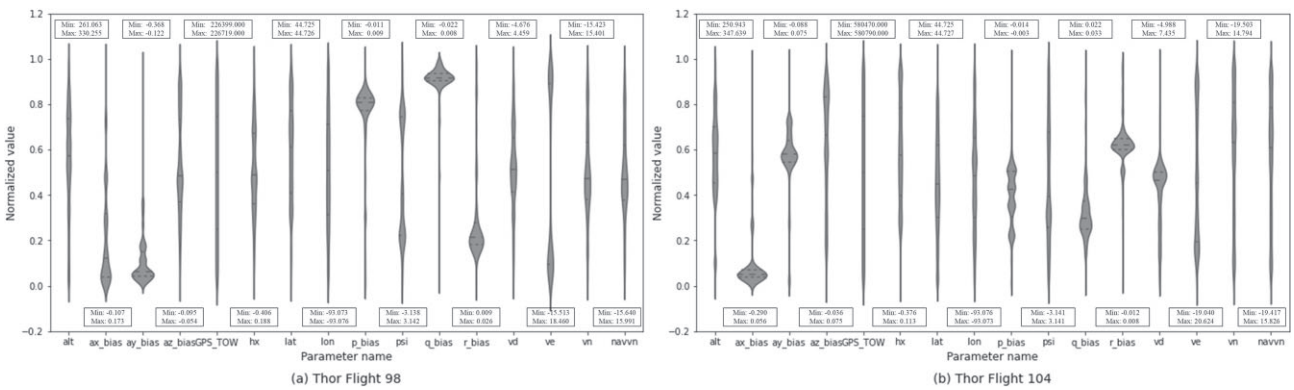

Figure 10: The violin plots of Thor Flight 98 and Thor Flight 104 datasets.

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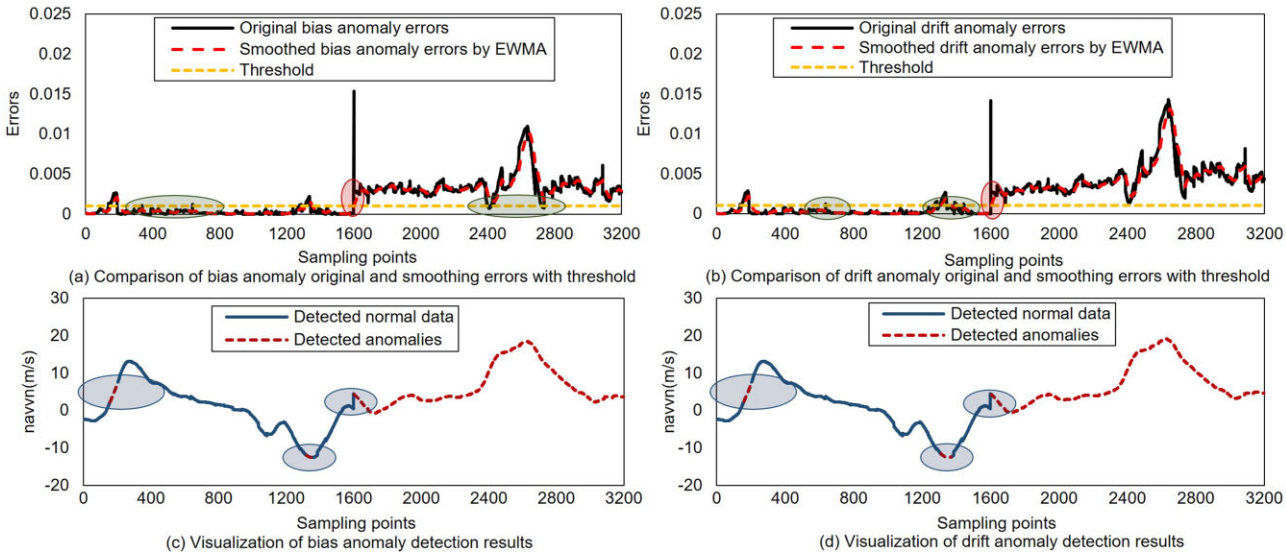

Figure 11: The anomaly detection results of CA-1DCL-EF for the bias and drift anomalies.

output values of the test samples are obtained by averaging or weighting the output values of these k samples. In this paper the value of k is taken as 50.

(iii) SVR: SVR predicts the values at the next moment by learning the features and patterns of the data. Anomalies are then identified by comparing the difference between the actual observed and predicted values. SVR can utilize various kernel functions, such as linear, polynomial, or radial basis functions, to learn the high-dimensional feature representation of the data. The kernel function in the SVR model in this paper is the radial basis function.

For the baseline methods, the models have an input size of n and an output size of 1, where n represents the number of parameters with correlation. In this study, n is 16.

4.4.3. *Experimental results on Thor Flight 98 dataset*

Anomaly detection

The anomaly detection visualization results of CA-1DCL-EF for the bias and drift anomalies are presented in Fig. 11. In Fig. 11a and b, the black, red, and orange curves represent the original errors, smoothed errors, and the anomaly determination threshold, respectively. Figure 11c and d depict the detected normal data and anomalies, respectively, with the blue and dark red curves. The original bias and drift anomaly errors are smoothed using EWMA, effectively reducing the false detection rate for both normal and abnormal data points, as indicated by the green circles in Fig. 11a and b. Although there are some missed and false detections, as shown by the blue circles in Fig. 11c and d, most data points are correctly detected. Specifically, CA-1DCL-EF correctly identifies 1522 normal and 1596 abnormal samples for the bias anomaly and 1496 normal and 1595 abnormal samples for the drift anomaly.

Table 3 lists the ACC, TPR, and FPR values of the baseline methods and CA-1DCL-EF. Although SVR correctly detects all the bias and drift anomaly samples with FPR values of 0.00%, its TPR values are relatively low for the bias and drift anomalies at 62.88% and 63.38%, respectively. This may be due to the relatively weak performance of SVR when dealing with complex, nonlinear relationships or noisy data. The overall anomaly detection performance of KNN is poor because of factors such as the selection of the k value and high data dimensionality, especially in terms of FPR values for the bias and drift anomalies, up to 28.88% and 21.13%, respectively. LSTM-RF and STC-LSTM perform relatively well compared with SVR and KNN, which is mainly because LSTM can effectively capture the long-term dependencies in flight data. The ACC and TPR values of LSTM-RF and STC-LSTM are more than 84.00% and 83.00%, respectively. Although the FPR values of CA-1DCL-EF are lower than those of SVR and LSTM-RF, CA-1DCL-EF achieves the highest ACC and TPR values with 97.44% and 95.13% for the bias anomaly, and 96.59% and 93.50% for the drift anomaly, respectively.

Table 3: Anomaly detection results of each method for the bias and drift anomalies.

Anomaly type	Methods	ACC	TPR	FPR
Bias anomaly	CA-1DCL-EF	**97.44%**	**95.13%**	0.25%
	KNN	74.19%	77.25%	28.88%
	SVR	81.44%	62.88%	**0.00%**
	STC-LSTM	84.22%	89.88%	21.44%
	LSTM-RF	93.34%	86.69%	**0.00%**
Drift anomaly	CA-1DCL-EF	**96.59%**	**93.50%**	0.33%
	KNN	78.06%	77.25%	21.13%
	SVR	81.69%	63.38%	**0.00%**
	STC-LSTM	89.03%	83.44%	5.68%
	LSTM-RF	93.72%	87.44%	**0.00%**

To further illustrate the advantages of the proposed method, Table 4 presents the difference values of each performance metric between CA-1DCL-EF and the baseline methods, which can be calculated by $\Delta\upsilon = \Delta\upsilon^{\text{our}} - \Delta\upsilon^{\text{me}}$, where Δ indicates the value of the difference, υ denotes the anomaly detection evaluation metrics and $\upsilon \in$ [ACC, TPR, FPR], *our* denotes CA-1DCL-EF, and *me* indicates the methods mentioned above and *me* $\in$ [SVR, KNN, STC-LSTM, LSTM-RF]. "+" indicates an increase and "−" indicates a decrease. It can be observed that the differences in ACC and TPR values between CA-1DCL-EF and the baseline methods are all positive, with respective ranges of variation being 2.87% to 23.25% and 5.25% to 32.25%. Although CA-1DCL-EF exhibits lower TPR values compared with SVR and LSTM-RF, as mentioned earlier, the differences are relatively small, with bias and drift anomalies of only

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Table 4: The differences between CA-1DCL-EF and the baseline methods.

Anomaly type	Method	ΔACC	ΔTPR	ΔFPR
Bias anomaly	KNN	+23.25%	+17.88%	−28.63%
	SVR	+16.00%	+32.25%	**+0.25%**
	STC-LSTM	+13.22%	+5.25%	−21.19%
	LSTM-RF	+4.10%	+8.44%	**+0.25%**
Drift anomaly	KNN	+18.53%	+16.25%	−20.80%
	SVR	+14.90%	+30.12%	**+0.33%**
	STC-LSTM	+7.56%	+10.06%	−5.35%
	LSTM-RF	+2.87%	+6.06%	**+0.33%**

Table 5: Data recovery results of the above methods.

Anomaly type	Method	MAE	MSE
Bias anomaly	CA-1DCL-EF	**0.04 107**	**0.00 214**
	KNN	0.09682	0.01641
	SVR	0.04184	0.00278
	STC-LSTM	0.04638	0.00333
	LSTM-RF	0.04744	0.00384
Drift anomaly	CA-1DCL-EF	0.04441	**0.00 239**
	KNN	0.09682	0.01641
	SVR	**0.03 805**	0.00245
	STC-LSTM	0.04632	0.00343
	LSTM-RF	0.04935	0.00417

0.25% and 0.33%, respectively. In contrast, the differences in the FPR values of KNN and STC-LSTM versus CA-1DCL-EF are more significantly higher at −28.63% and −21.19%, and −20.80% and −5.35% for the bias and drift anomalies, respectively. The results show that CA-1DCL-EF can significantly improve in most metrics compared with the benchmark methods, verifying the effectiveness of CA-1DCL-EF in detecting different anomaly types.

Data recovery

Figure 12 illustrates the visualization of data recovery results for CA-1DCL-EF, KNN, SVR, STC-LSTM, and LSTM-RF, where the black and red curves represent the original and estimated data, respectively. KNN and SVR show more pronounced recovery biases in some local regions, as shown by the orange and blue circles in Fig. 12. This may be due to the sensitivity of KNN to noise or local fluctuations and the lack of flexibility in SVR to capture certain complex nonlinear relationships, resulting in more pronounced restoration biases in some local regions. In contrast, the predictions of CA-1DCL-EF, STC-LSTM, and LSTM-RF do not show any significant biases overall. This suggests that these methods are more successful in capturing the nonlinear relationships and patterns within the data.

Table 5 lists the MAE and MSE values of the regression performance metrics for the above methods. KNN exhibits the highest MAE and MSE values, with 0.09 682 and 0.01 641, respectively, for the bias and drift anomalies. SVR, STC-LSTM, LSTM-RF, and CA-1DCL-EF present lower MAE and MSE values for the bias and drift anomalies. Specifically, for the drift anomaly, CA-1DCL-EF has an MAE value of 0.04 441, which is slightly lower than that of SVR, but compared with SVR, STC-LSTM, and LSTM-RF, it achieves the lowest MSE value of 0.00 239. Particularly for the bias anomaly, CA-1DCL-EF exhibits the lowest MAE and MSE values of 0.04 107 and 0.00 214, respectively, compared with the baseline methods. In addition, it is observed that for the drift anomaly, the MAE and MSE

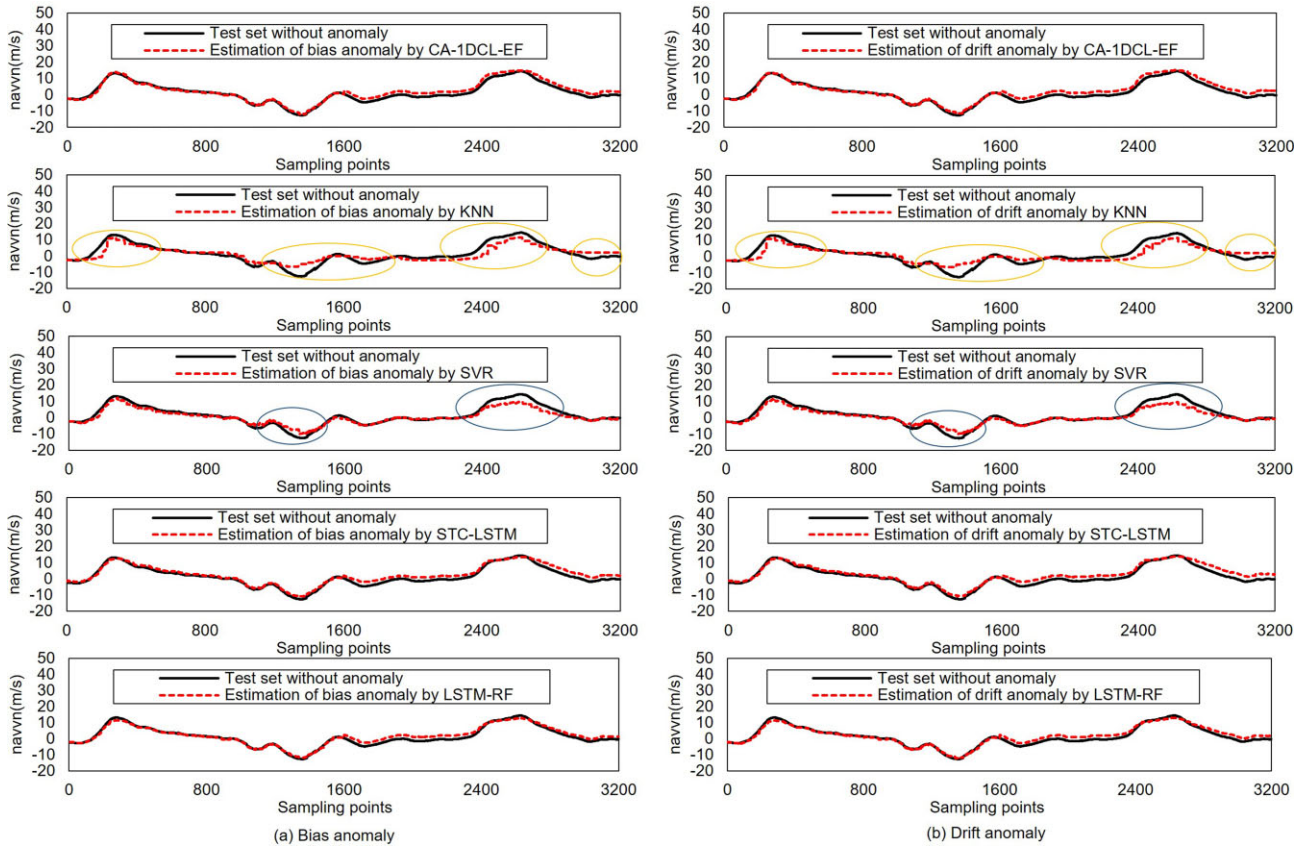

Figure 12: Data recovery visualization of the above methods.

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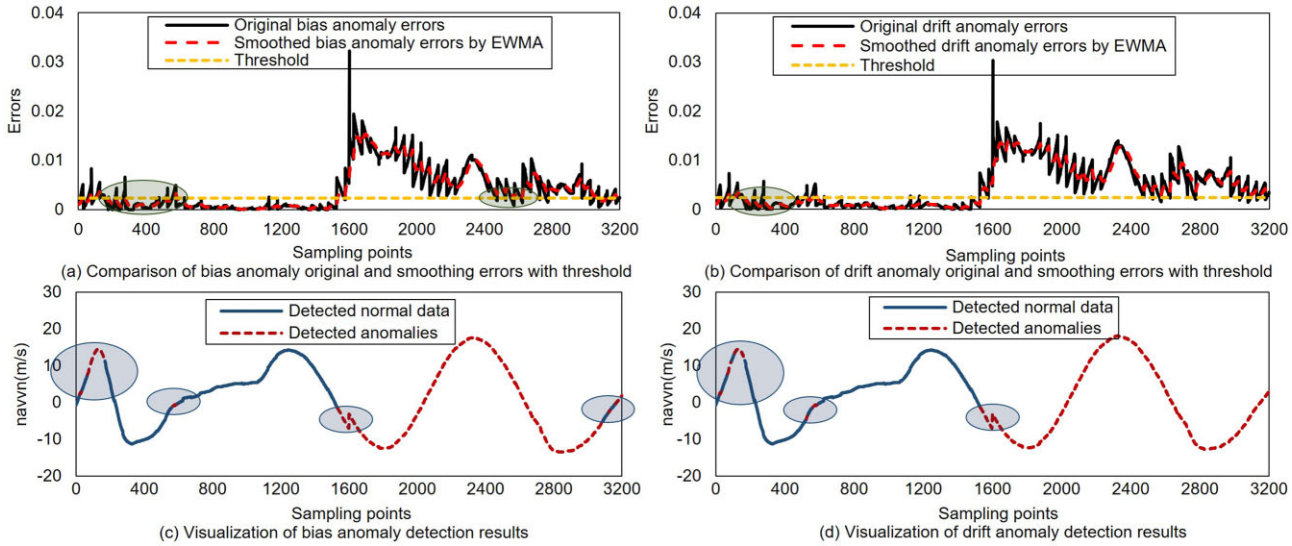

Figure 13: The anomaly detection results of CA-1DCL-EF for the bias and drift anomalies.

Table 6: Anomaly detection results of each method for the bias and drift anomalies.

Anomaly type	Methods	ACC	TPR	FPR
Bias anomaly	CA-1DCL-EF	**91.53%**	**87.50%**	**4.44%**
	KNN	64.59%	57.88%	30.94%
	SVR	65.09%	57.19%	27.00%
	STC-LSTM	85.41%	78.25%	7.44%
	LSTM-RF	81.50%	70.50%	7.50%
Drift anomaly	CA-1DCL-EF	**94.22%**	**88.44%**	**0.00%**
	KNN	69.25%	57.88%	19.38%
	SVR	71.31%	57.31%	14.69%
	STC-LSTM	87.34%	83.06%	8.38%
	LSTM-RF	80.31%	60.63%	**0.00%**

Table 7: The differences between CA-1DCL-EF and the baseline methods.

Anomaly type	Method	ΔACC	ΔTPR	ΔFPR
Bias anomaly	KNN	+26.94%	+29.62%	−26.50%
	SVR	+26.44%	+30.30%	−22.56%
	STC-LSTM	+6.12%	+9.25%	−3.00%
	LSTM-RF	+10.03%	+17.00%	−3.06%
Drift anomaly	KNN	+24.97%	+30.56%	−19.38%
	SVR	+22.91%	+31.13%	−14.69%
	STC-LSTM	+6.88%	+5.38%	−8.38%
	LSTM-RF	+13.91%	+27.81%	**+0.00%**

values of the above methods are generally higher than those of the bias anomaly. This may be due to the greater degree of injected drift anomaly than the bias anomaly, which makes the data deviate more significantly from the norm, thus triggering more significant uncertainty and volatility. With this uncertainty, the model's data recovery ability may be more challenged, leading to relatively high MAE and MSE values. Nevertheless, the overall data recovery performance of CA-1DCL-EF is still significantly better than that of the other compared methods, further validating its effectiveness in data recovery.

4.4.4. *Experimental results on Thor Flight 104 dataset*

Anomaly detection

Figure 13 shows the results of the anomaly detection visualization of CA-1DCL-EF for the bias and drift anomalies. It can be seen that using EWMA for error smoothing still significantly reduces the false detection rate of certain points with drastic error variations, as shown by the green circles in Fig. 13. However, it is undeniable that CA-1DCL-EF appears to have a relatively high number of missed or false detections, as shown by the black circles in Fig. 13.

Table 6 lists the anomaly detection performance metrics for the above methods. It can be clearly observed that CA-1DCL-EF significantly outperforms the other compared methods in bias and drift anomaly detection. Specifically, the ACC, TPR, and FPR values for CA-1DCL-EF are 91.53%, 87.50%, and 4.44% for the bias anomaly, and 94.22%, 88.44%, and 0.00% for the drift anomaly, respectively. The overall performances of KNN and SVR in anomaly detection are relatively poor, especially with high FPR values of 30.94% and 27.00%, and 19.38% and 14.69% for the bias and drift anomalies, respectively. In contrast, STC-LSTM and LSTM-RF exhibit relatively lower FPR values of 7.44% and 7.50%, and 8.38% and 0.00%, for the bias and drift anomalies, respectively. However, STC-LSTM and LSTM-RF still perform relatively poorly in terms of ACC and TPR values.

Similarly, to provide more insight into the advantages of the proposed method, the difference values of the performance metrics between CA-1DCL-EF and the baseline methods are demonstrated in Table 7. As described previously, the overall anomaly detection performances of KNN and SVR are relatively poor, which can also be reflected in their ΔACC, ΔTPR, and ΔFPR values. For example, for the bias and drift anomalies, the ACC and TPR values of CA-1DCL-EF are higher than those of KNN by 26.94% and 29.62%, and 24.97% and 30.56%, respectively. The FPR values of KNN are 26.50% and 19.38% higher than those of CA-1DCL-EF, respectively, for the bias and drift anomalies. For the bias and drift anomalies, the differences between SVR and CA-1DCL-EF also show a similar trend to KNN. Compared with KNN and SVR, although the performance differences between STC-LSTM and LSTM-RF versus CA-1DCL-EF are reduced, the magnitude of the difference change is still relatively high for most performance metrics.

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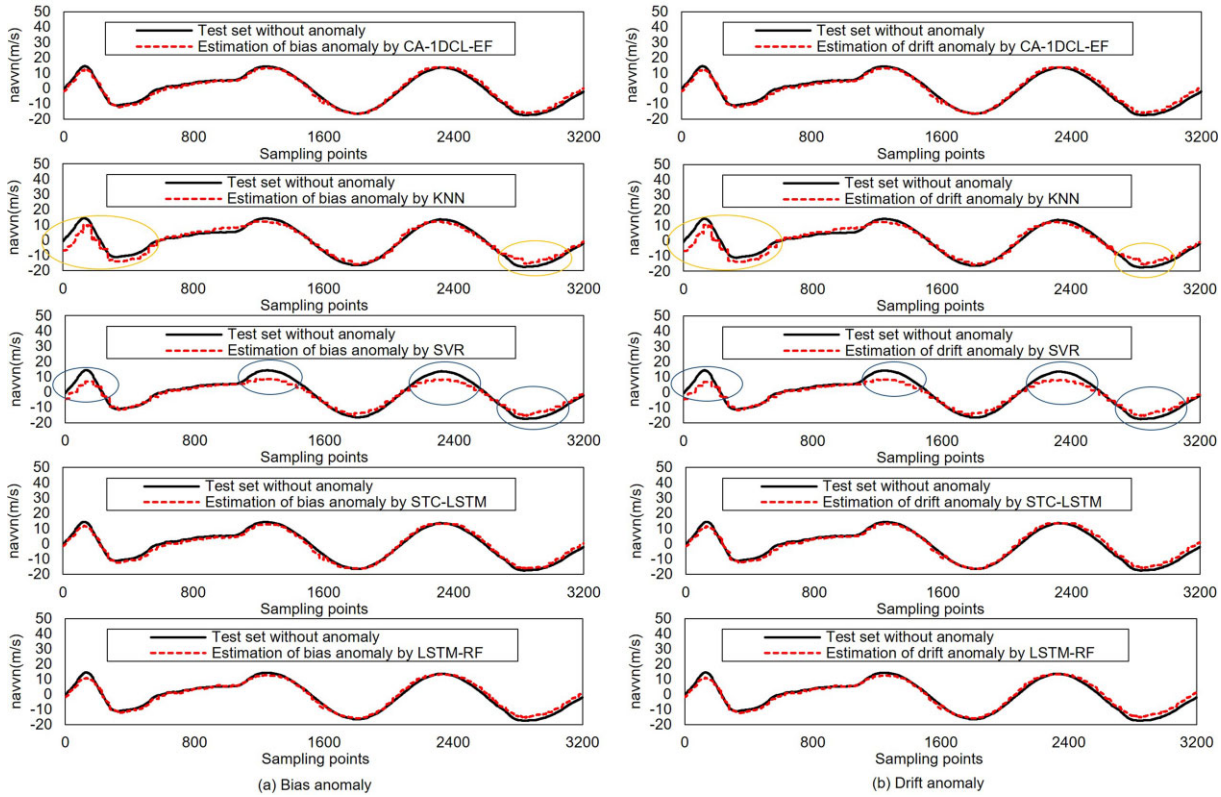

Figure 14: Data recovery visualization of the above methods.

Table 8: Data recovery results of the above methods.

Anomaly type	Method	MAE	MSE
Bias anomaly	CA-1DCL-EF	**0.03 574**	**0.00 183**
	KNN	0.06 126	0.00 711
	SVR	0.05 506	0.00 483
	STC-LSTM	0.04 061	0.00 251
	LSTM-RF	0.03 961	0.00 222
Drift anomaly	CA-1DCL-EF	**0.04 049**	**0.00 226**
	KNN	0.06 126	0.00 711
	SVR	0.05 526	0.00 487
	STC-LSTM	0.04 158	0.00 260
	LSTM-RF	0.04 211	0.00 250

Table 9: Anomaly detection results of LSTM and 1D CNN.

Dataset name	Anomaly type	Method	ACC	TPR	FPR
Thor Flight 98	Bias anomaly	LSTM	80.22%	60.44%	**0.00%**
		1D CNN	87.69%	85.56%	10.19%
	Drift anomaly	LSTM	79.72%	59.44%	**0.00%**
		1D CNN	92.66%	85.69%	0.38%
Thor Flight 104	Bias anomaly	LSTM	89.81%	**89.81%**	10.19%
		1D CNN	67.16%	38.06%	3.75%
	Drift anomaly	LSTM	93.84%	**90.00%**	2.31%
		1D CNN	68.96%	37.94%	**0.00%**

Data recovery

Figure 14 shows the visualization of data recovery results for the above methods. It can be clearly seen that KNN and SVR are still unable to accurately predict some local region data points well, as shown by the orange and blue circles in Fig. 14, for the reasons mentioned earlier. Compared with KNN and SVR, LSTM-RF, STC-LSTM, and CA-1DCL-EF do not exhibit more significant recovery biases in predicting different anomaly types of data.

Table 8 lists the MAE and MSE values for the above methods. For the baseline methods, KNN and SVR show significantly higher MAE and MSE values for the bias and drift anomalies than STC-LSTM, and LSTM-RF. Taking the bias anomaly as an example, the MAE and MSE values of STC-LSTM and LSTM-RF are 0.04 061 and 0.00 251, and 0.03 961 and 0.00 222, respectively, whereas those of KNN and SVR are 0.06 126 and 0.00 711, and 0.05 506 and 0.00 483 respectively. Compared with the baseline methods, CA-1DCL-EF achieves the optimal data recovery performance, i.e., the smallest MAE and MSE values of 0.03 574 and 0.00 183, and 0.04 049 and 0.00 226, respectively, for the bias and drift anomalies. Similarly, the MAE and MSE values of the above methods for the drift anomaly are generally higher than those for the bias anomaly, which is due to the reasons mentioned previously.

Ablation study

This subsection investigates the anomaly detection and data recovery performance of LSTM and 1D CNN as ablation experiments on different flight datasets. The anomaly detection performance metrics of LSTM and 1D CNN on different datasets are given in Table 9. It can be seen that LSTM can detect all the bias and drift anomaly samples of the Thor Flight 98 dataset, i.e., all the FPR values are 0.00%, but its ACC and TPR values are relatively low. 1D CNN shows relatively good anomaly detection performance on the Thor Flight 98 dataset, with its ACC and TPR values exceeding 87.00% and 85.00% for the bias and drift anomalies, respectively. In contrast, for the Thor Flight 104 dataset, LSTM performs better in terms of TPR values, even exceeding CA-1DCL-EF, for the bias and drift anomalies of 89.81%

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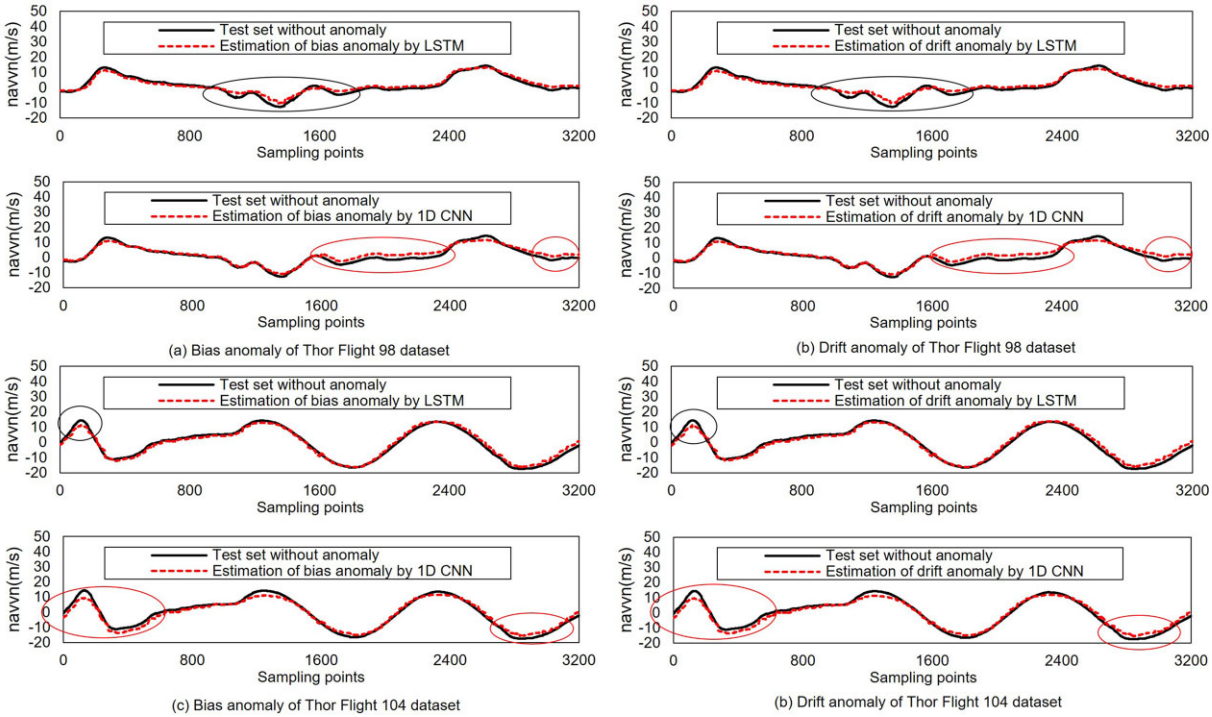

Figure 15: Data recovery visualization of LSTM and 1D CNN.

Table 10: Data recovery results of LSTM and 1D CNN.

Dataset name	Anomaly type	Method	MAE	MSE
Thor Flight 98	Bias anomaly	LSTM	**0.04 096**	**0.00 223**
		1D CNN	0.07 214	0.00 736
	Drift anomaly	LSTM	**0.04 161**	**0.00 227**
		1D CNN	0.07 202	0.00 756
Thor Flight 104	Bias anomaly	LSTM	0.03 834	0.00 205
		1D CNN	0.04 355	0.00 304
	Drift anomaly	LSTM	0.04 151	0.00 237
		1D CNN	0.04 449	0.00 320

and 90.00%, respectively, but still shows high FPR values of 10.19% and 2.31%, respectively. Although 1D CNN maintains low FPR values on the Thor Flight 104 dataset, its ACC and TPR values perform unsatisfactorily. This phenomenon may be due to the difference in data distribution between different datasets. Although LSTM and 1D CNN have their own advantages and disadvantages on different datasets, they have a synergistic effect in reducing the FPR values or improving the ACC and TPR values. This synergy allows CA-1DCL-EF to achieve a better balance in anomaly detection performance and provide a more reliable solution.

Figure 15 shows the recovery visualization of LSTM and 1D CNN on different flight datasets. It can be clearly seen that LSTM and 1D CNN show some recovery biases on different flight datasets, as shown by the red and black circles in Fig. 15. To quantify the recovery performance of LSTM and 1D CNN, Table 10 lists their MAE and MSE values on different flight datasets. An obvious phenomenon is that the MAE and MSE of 1D CNN are generally higher than those of LSTM for different flight datasets, which can also be seen from the red circles in Fig. 15. For example, for the bias and drift anomalies of the Thor Flight 104 dataset, the MAE and MSE values of 1D CNN are 0.04 355 and 0.00 304, and 0.04 449 and 0.00 320, whereas those of LSTM are 0.03 834 and 0.00 205, and 0.04 151 and 0.00 237, respectively. The difference in performance between LSTM and 1D CNN may be due to the temporal nature of the flight data, whereas LSTM is more advantageous relative to 1D CNN in processing this type of data. Meanwhile, it is observed that the MAE and MSE values of the bias anomaly are generally lower than the drift anomaly for 1D CNN and LSTM on all flight datasets, for the reasons described previously. In addition, for the Thor Flight 98 dataset, LSTM outperforms CA-1DCL-EF with MAE and MSE values of 0.04 096 and 0.00 223, and 0.04 161 and 0.00 227, respectively, for the bias and drift anomalies. However, LSTM focuses on long-term dependency modeling and cannot accurately predict local regions with sharp changes, as shown by the black circles in Fig. 15. This leads to more false detections. As mentioned earlier, LSTM has relatively low TPR values despite being able to correctly detect all anomalous samples in the Thor Flight 98 dataset.

4.4.5. *Effects of parameter selection and EF*

To assess the impact of the input parameter number and EF on the model's performance, Table 11 presents the anomaly detection results of 1DCL-EF and CA-1DCL on different flight datasets. 1DCL-EF represents the model without parameter selection and CA-1DCL denotes the model without EF. It can be seen that CA-1DCL has relatively good anomaly detection performance on different flight datasets. For the bias and drift anomalies of the Thor Flight 98 dataset, the ACC, TPR, and FPR values of CA-1DCL are 96.13%, 94.44%, and 2.19%, and 96.38%, 92.75%, and 0.06%, respectively. It is noteworthy that the FPR value of CA-1DCL for the drift anomaly is better than that of CA-1DCL-EF. This is because EWMA is not sensitive to rapid changes in data and is more concerned with the overall trend, which may lead to abnormal starting position data being incorrectly judged as normal data, as shown by the red circles in Figs. 11a and b. This is not always the case, however, and is related to the threshold and errors. Although for the Thor

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Table 11: Anomaly detection results of CA-1DCL and 1DCL-EF.

Dataset name	Anomaly type	Method	ACC	TPR	FPR
Thor Flight 98	Bias anomaly	CA-1DCL	96.13%	94.44%	2.19%
		1DCL-EF	84.84%	83.19%	13.50%
	Drift anomaly	CA-1DCL	96.38%	92.75%	**0.06%**
		1DCL-EF	80.53%	70.06%	7.13%
Thor Flight 104	Bias anomaly	CA-1DCL	88.06%	85.63%	9.50%
		1DCL-EF	88.28%	86.88%	10.31%
	Drift anomaly	CA-1DCL	90.94%	85.57%	3.69%
		1DCL-EF	85.69%	71.38%	**0.00%**

Table 12: Anomaly detection results of PCC-1DCL-EF.

Dataset name	Anomaly type	ACC	TPR	FPR
Thor Flight 98	Bias anomaly	94.56%	90.56%	1.44%
	Drift anomaly	94.25%	88.63%	**0.13%**
Thor Flight 104	Bias anomaly	80.19%	62.50%	**2.13%**
	Drift anomaly	79.19%	58.94%	0.56%

Flight 98 dataset, the difference in performance between CA-1DCL and CA-1DCL-EF is relatively small, however, for the Thor Flight 104 dataset, the advantage of error smoothing is much more pronounced, especially in reducing the FPR values.

For 1DCL-EF, its overall anomaly detection performance on different flight datasets is relatively poor. For the bias and drift anomalies of the Thor Flight 98 dataset, the ACC, TPR, and FPR values for 1DCL-EF are 84.84%, 83.19%, and 13.50%, and 80.53%, 70.06%, and 7.13%, respectively. For the Thor Flight 104 dataset, although 1DCL-EF achieves an FPR value of 0.00% for the drift anomaly, it still performs relatively poorly for ACC and TPR values and in the bias anomaly detection. This may be due to too many input parameters, resulting in an underfitting of 1DCL-EF. In contrast, CA-1DCL-EF improves the anomaly detection performance by effectively overcoming the underfitting problem through parameter selection.

4.4.6. *Model performance of PCC-based parameter selection*

To further explore the advantage of MIC-based parameter selection, Table 12 shows the anomaly detection results of the method using PCC for parameter selection, named PCC-1DCL-EF, on different flight datasets. PCC-1DCL-EF performs well on the Thor Flight 98 dataset, with ACC, TPR, and FPR values of 94.56%, 90.56%, and 1.44%, and 94.25%, 88.63%, and 0.13% for the bias and drift anomalies, respectively. However, for the Thor Flight 104 dataset, it suffers from a more pronounced false detection problem, with TPR values of only 62.50% and 58.94% for the bias and drift anomalies, respectively. Although PCC-1DCL-EF has relatively low FPR values, in some cases even lower than CA-1DCL-EF, its overall performance is still inferior to CA-1DCL-EF. This difference is especially significant on the Thor Flight 104 dataset. The reasons for the relatively significant difference in the performance of PCC-1DCL-EF on different flight datasets may be attributed to multiple factors, such as differences in data distribution and the loss of key nonlinear parameters that positively impact model performance.

Figure 16 shows the data recovery visualization of PCC-1DCL-EF on different flight datasets. It can be seen that PCC-1DCL-EF does not show significant recovery biases on different flight datasets. Table 13 quantifies the recovery performance metrics of PCC-1DCL-EF on different flight datasets. It can be clearly observed that the MAE values of PCC-1DCL-EF are lower than those of CA-1DCL-EF for different flight datasets, while the MSE values are higher than those of CA-1DCL-EF. This is due to the fact that the MIC-based parameter selection contains many nonlinear correlation parameters which may lead to large absolute errors on some samples, making the MAE values relatively high. Therefore, compared with PCC-1DCL-EF, CA-1DCL-EF achieves a better trade-off between anomaly detection and data recovery performance by taking into account the nonlinear correlation parameters in the flight data.

4.4.7. *Discussion*

This subsection discusses the research results on solving the problems of random noise, parameter selection, and feature extraction using the proposed framework CA-1DCL-EF and gives the following main findings:

First, compared with the baseline methods, CA-1DCL-EF achieves excellent anomaly detection performance on different flight datasets with the highest TPR and ACC values while maintaining lower FPR values. Furthermore, CA-1DCL-EF also exhibits superior recovery performance on different flight datasets, which can be reflected by its lower MAE and MSE values. These results demonstrate that CA-1DCL-EF has good generalization and robustness, providing an effective and practical solution to ensure the safe flight of UAVs and their intelligent operation and maintenance.

Second, the ablation experiments conducted in this study provide valuable insights into the individual contributions of 1D CNN and LSTM in anomaly detection and data recovery tasks. The results reveal that using 1D CNN or LSTM alone yielded relatively poor temporal feature modeling or local feature extraction performance. However, integrating these two approaches in the designed 1D CNN-LSTM model synergizes their strengths and better balances data recovery and anomaly detection performance. This finding highlights the importance of utilizing hybrid models that leverage multiple deep-learning architectures to handle the complex nature of sequential data effectively.

Third, by utilizing the feature extraction capability of 1D CNN-LSTM model and combining the EWMA and MIC methods to deal with noise and capture complex spatio-temporal correlation, the anomaly detection and data recovery impacts of CA-1DCL-EF are improved more significantly. This integrated approach provides a valuable reference for the practical application of accurate and reliable UAV anomaly detection and data recovery. Meanwhile, the overall better performance of anomaly detection and data recovery based on MIC parameter selection compared with PCC provides important guidance for future research and applications.

Although this study reveals some important findings, some limitations must be acknowledged. First, UAV flight data may be affected by multiple factors, resulting in different types and degrees of anomalies. Manually injected anomaly data are often

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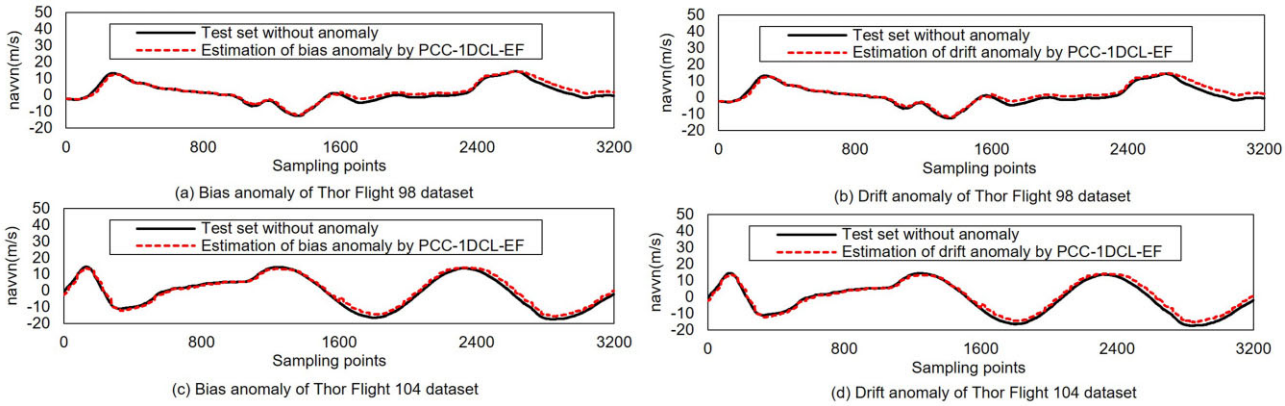

Figure 16: Data recovery visualization of PCC-1DCL-EF.

Table 13: Data recovery results of PCC-1DCL-EF.

Dataset name	Anomaly type	MAE	MSE
Thor Flight 98	Bias anomaly	**0.03 867**	0.00254
	Drift anomaly	**0.04 106**	0.00284
Thor Flight 104	Bias anomaly	**0.03 520**	0.00199
	Drift anomaly	**0.03 804**	0.00228

pre-defined fixed patterns that cannot fully simulate the complex anomaly types in actual UAV flight data. Second, given the limitations of the data employed in this paper, the assessment of noise's impact on anomaly detection performance remains incomplete. Third, the threshold setting in this paper is calculated using a statistical method and is static. It cannot be adaptively adjusted according to the dynamic changes of UAV flight data, which may lead to more false alarms or missed alarms. Finally, although this paper uses the advanced methods in the field as the comparison methods, they are not sufficient.

Therefore, future research can focus on the following aspects to address the aforementioned limitations: (i) Incorporating more comprehensive anomaly types, such as stuck anomaly and compound anomaly (Y. He *et al.*, 2018; Zhong *et al.*, 2022), into future experiments to improve the representativeness and persuasiveness of the experiments. (ii) Introducing different noise levels to assess their impact on the accuracy and stability of the model. (iii) Combining adaptive methods (Fang *et al.*, 2022; Hou *et al.*, 2023) for anomaly detection to reduce the false alarm rate and enhance the generalization performance of the model. (iv) Including more advanced anomaly detection methods for time-series data into the study, such as TimesNet (Wu *et al.*, 2022), to fully evaluate the effectiveness of the model.

5. Conclusions

To achieve accurate anomaly detection and reliable data recovery in UAV flight data, this paper proposed a novel comprehensive framework named CA-1DCL-EF. First, the MIC-based method is used to select correlation parameters as model input to avoid the negative influence of unrelated variables on the model. Second, a regression model based on 1D CNN-LSTM was designed to comprehensively capture the spatio-temporal features in the flight data and realize more accurate parameter mapping. Then, the errors were smoothed by the EWMA method to reduce the impact of random noise on the model performance. Finally, experiments were performed on real UAV flight datasets injected with bias and drift anomalies. The anomaly detection and data recovery results on different flight datasets demonstrated that the overall performance of CA-1DCL-EF was significantly better than the baseline methods. In addition, the necessity and effectiveness of parameter selection and error smoothing on model performance improvement were verified, offering reference value for UAV flight data anomaly detection and recovery.

In the future, the effective integration of various methods will be continuously optimized and explored for practical problems to further improve the anomaly detection performance of the model. Meanwhile, different anomaly types, noise levels, and other advanced methods will also be included in future research and experiments to further verify the generality and effectiveness of the proposed method. In addition, research will be conducted on the adaptive detection problem to develop methods that can dynamically adjust the anomaly detection thresholds, thereby reducing the false alarm rate.

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Conflict of interest statement

None declared.

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